Stability-Guided Pixel-Wise Correlation (SG-PWC) Beamforming for Clutter Suppression in Echocardiography

Background, Motivation, and Objective

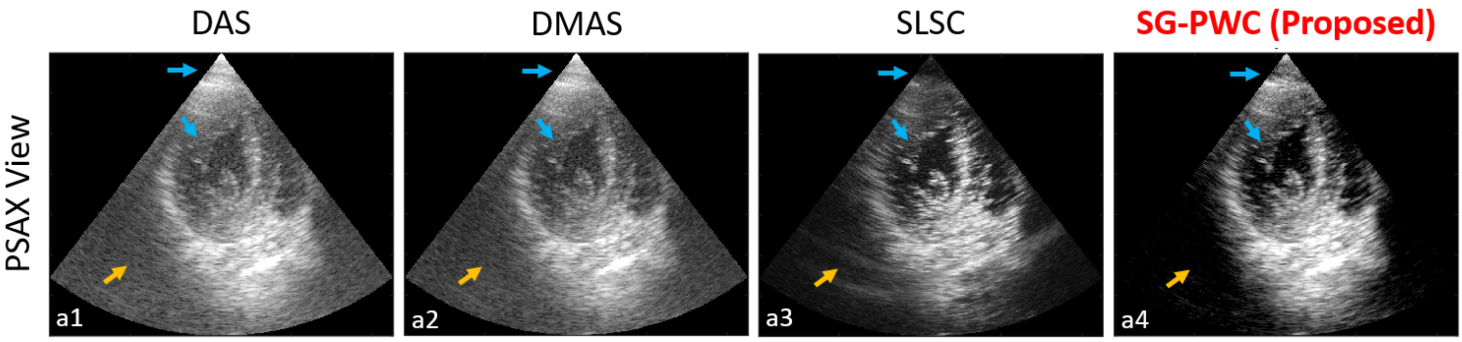
Ultrasound cardiac imaging is frequently degraded by acoustic clutter and incoherent echoes that obscure anatomical structures and reduce diagnostic visibility. Conventional delay-and-sum (DAS) beamforming preserves speckle texture but is highly susceptible to clutter. Nonlinear and coherence-based approaches such as delay-multiply-and-sum (DMAS) and short-lag spatial coherence (SLSC) can improve cavity contrast but often introduce speckle distortion or excessive smoothing that reduces structural detail. To address these limitations, we propose Stability-Guided Pixel-Wise Correlation beamforming (SCB), an approach that enhances coherent tissue responses by evaluating the stability of spatial correlations across receive channels.

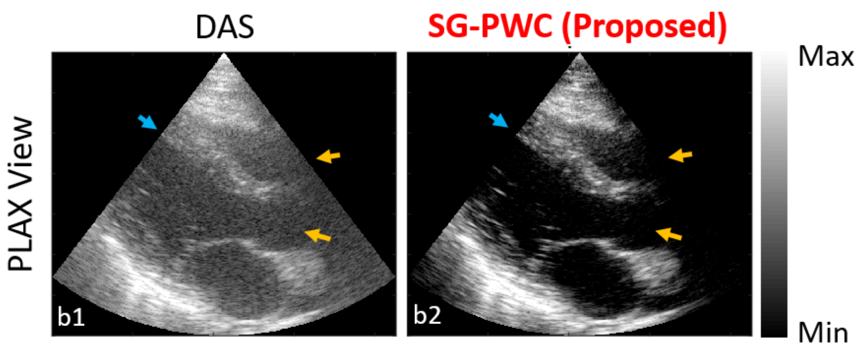
Statement of Contribution/Methods

For each image pixel, normalized cross-correlations between delayed channel signals are computed over multiple short spatial lags. Instead of relying only on the correlation magnitude, the proposed method analyzes the dispersion of lag-wise correlations around their mean value. A stability-guided weighting factor is then derived by penalizing large deviations of individual lag correlations from the mean correlation. Echoes exhibiting consistent correlation across lags produce a strong stability factor, while unstable or incoherent signals associated with clutter and reverberation are suppressed. This stability factor is applied pixel-wise to the beamformed signal, reinforcing reliable coherent echoes while attenuating unstable contributions.

Results/Discussion

The method was evaluated on in-vivo cardiac ultrasound data and compared with DAS, DMAS, and SLSC beamformers in parasternal short-axis (PSAX) and parasternal long-axis (PLAX) views. Qualitative results show substantially cleaner ventricular cavities, sharper myocardial borders, and improved delineation of cardiac structures while preserving natural speckle texture. Quantitative analysis demonstrates that SG-PWC achieves the highest contrast performance with a gCNR of 1.00 and a CNR of 4.2, outperforming SLSC (gCNR 0.99, CNR 3.2) and both DAS and DMAS (gCNR 0.96, CNR 2.5). These results indicate that incorporating correlation stability provides robust clutter suppression while maintaining structural fidelity in cardiac ultrasound imaging.





Quantitative Values

Method	gCNR	CNR
DAS	0.96	2.5
DMAS	0.96	2.5
SLSC	0.99	3.2
SG-PWC	**1.00**	**4.2**